# Secure Sight Statistical Analysis Report

Total runs: 25

Total batches: 5

# 1. Global Metrics

| Metric | Mean | Std Dev | Variance | Min | Max | Median | Count |
|---|---|---|---|---|---|---|---|
| **accuracy** | 0.882542 | 0.0193645576035860 6 | 0.000360 | 0.848468 | 0.915821 | 0.884026 | 25 |
| **precision** | 0.872027 | 0.0255890782745505 36 | 0.000629 | 0.827591 | 0.928699 | 0.876727 | 25 |
| **recall** | 0.883841 | 0.0255729570302154 8 | 0.000628 | 0.837560 | 0.929544 | 0.886608 | 25 |
| **f1_score** | 0.875067 | 0.0205883343522711 84 | 0.000407 | 0.836546 | 0.914991 | 0.873292 | 25 |
| **auc** | 0.895226 | 0.0278325949631116 1 | 0.000744 | 0.842868 | 0.948310 | 0.899564 | 25 |
| **latency** | 9.411680 | 1.4892998067497136 | 2.129293 | 7.755605 | 12.088238 | 9.603186 | 25 |

# 2. Confidence Intervals

| Metric | Mean | CI Lower | CI Upper |
|---|---|---|---|
| **accuracy** | 0.882542 | 0.874548 | 0.890535 |
| **precision** | 0.872027 | 0.861464 | 0.882590 |
| **recall** | 0.883841 | 0.873285 | 0.894397 |
| **f1_score** | 0.875067 | 0.866569 | 0.883566 |
| **auc** | 0.895226 | 0.883737 | 0.906715 |
| **latency** | 9.411680 | 8.796927 | 10.026433 |

# 3. Per-Batch Metrics

## Batch 1

| Metric | Mean | Std Dev | Variance | Min | Max | Median |
|---|---|---|---|---|---|---|
| **accuracy** | 0.886268 | 0.0119444215754667 | 0.000114 | 0.868468 | 0.895821 | 0.891550 |
| **precision** | 0.872166 | 0.0246193157340565 23 | 0.000485 | 0.847591 | 0.908699 | 0.876727 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **recall** | 0.892712 | 0.021994546 07605115 | 0.000387 | 0.857560 | 0.909544 | 0.90156 9 |
| **f1_score** | 0.882289 | 0.013101402 848217064 | 0.000137 | 0.860718 | 0.892784 | 0.88321 6 |
| **auc** | 0.898001 | 0.018620549 397197966 | 0.000277 | 0.868274 | 0.911645 | 0.90956 4 |
| **latency** | 7.942580 | 0.124279079 4006885 | 0.012356 | 7.755605 | 8.069188 | 8.00265 5 |

### Batch 2

| Metric | Mean | Std Dev | Variance | Min | Max | Median |
|---|---|---|---|---|---|---|
| **accuracy** | 0.860797 | 0.012714226 039313771 | 0.000129 | 0.848468 | 0.875821 | 0.85967 8 |
| **precision** | 0.847039 | 0.026059802 354930654 | 0.000543 | 0.827591 | 0.888699 | 0.83108 8 |
| **recall** | 0.862315 | 0.024222705 933271325 | 0.000469 | 0.837560 | 0.889544 | 0.86534 4 |
| **f1_score** | 0.855875 | 0.014457348 164581853 | 0.000167 | 0.840718 | 0.872784 | 0.86194 0 |
| **auc** | 0.869743 | 0.021188960 676254497 | 0.000359 | 0.848274 | 0.891645 | 0.87095 7 |
| **latency** | 7.955887 | 0.135384970 29374835 | 0.014663 | 7.755605 | 8.069188 | 8.00265 5 |

### Batch 3

| Metric | Mean | Std Dev | Variance | Min | Max | Median |
|---|---|---|---|---|---|---|
| **accuracy** | 0.902635 | 0.016385511 72395041 | 0.000215 | 0.881516 | 0.915821 | 0.91155 0 |
| **precision** | 0.894962 | 0.021758254 725068156 | 0.000379 | 0.871088 | 0.928699 | 0.89672 7 |
| **recall** | 0.902107 | 0.025812732 31739681 | 0.000533 | 0.872475 | 0.929544 | 0.90938 6 |
| **f1_score** | 0.895396 | 0.021076724 004960655 | 0.000355 | 0.866546 | 0.914991 | 0.90194 0 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **auc** | 0.914132 | 0.031954303662711785 | 0.000817 | 0.872868 | 0.948310 | 0.929564 |
| **latency** | 9.634360 | 0.20516477632479757 | 0.033674 | 9.306726 | 9.858685 | 9.683026 |

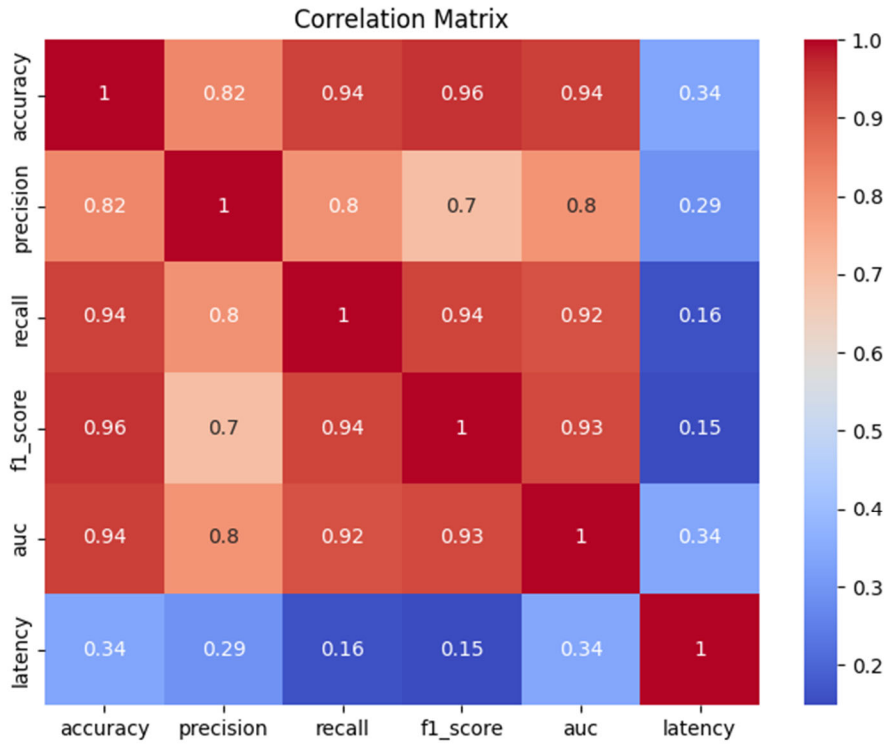
### Batch 4

| Metric | Mean | Std Dev | Variance | Min | Max | Median |
|---|---|---|---|---|---|---|
| **accuracy** | 0.872635 | 0.016385511 72395041 | 0.000215 | 0.851516 | 0.885821 | 0.881550 |
| **precision** | 0.864962 | 0.021758254 72506816 | 0.000379 | 0.841088 | 0.898699 | 0.866727 |
| **recall** | 0.872107 | 0.025812732 31739681 | 0.000533 | 0.842475 | 0.899544 | 0.879386 |
| **f1_score** | 0.865396 | 0.021076724 004960655 | 0.000355 | 0.836546 | 0.884991 | 0.871940 |
| **auc** | 0.884132 | 0.031954303 662711785 | 0.000817 | 0.842868 | 0.918310 | 0.899564 |
| **latency** | 9.634360 | 0.20516477632479757 | 0.033674 | 9.306726 | 9.858685 | 9.683026 |

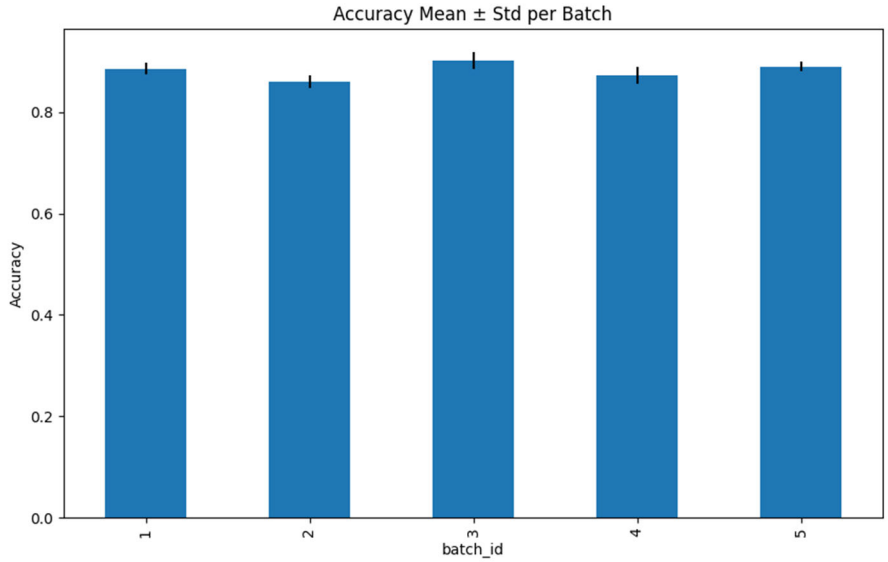
### Batch 5

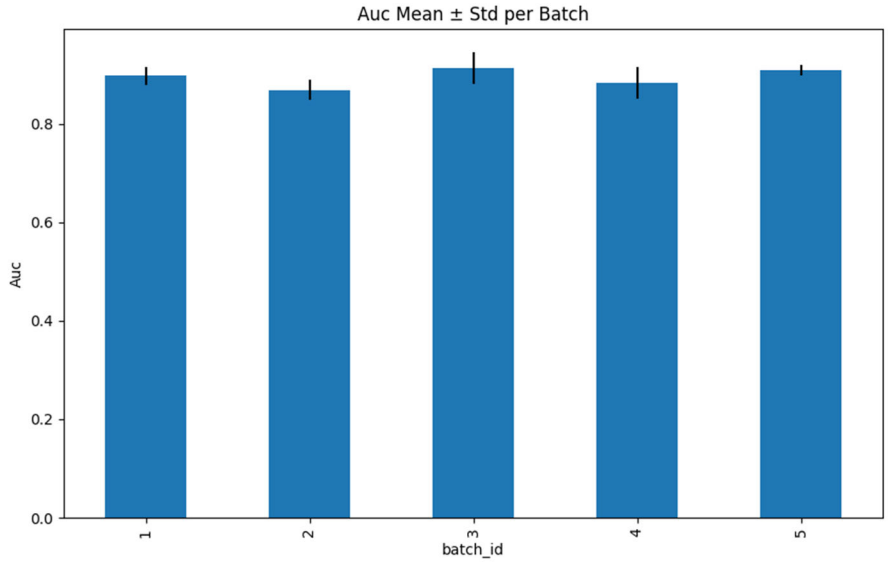
| Metric | Mean | Std Dev | Variance | Min | Max | Median |
|---|---|---|---|---|---|---|
| **accuracy** | 0.890374 | 0.009808132 162759992 | 0.000077 | 0.881704 | 0.901695 | 0.884026 |
| **precision** | 0.881007 | 0.009771394 653209288 | 0.000076 | 0.864985 | 0.889313 | 0.881116 |
| **Recall** | 0.889965 | 0.014392742 289228875 | 0.000166 | 0.869431 | 0.906155 | 0.886608 |
| **f1_score** | 0.876381 | 0.010626368 899567607 | 0.000090 | 0.862733 | 0.891283 | 0.873292 |
| **Auc** | 0.910123 | 0.010519467 412790997 | 0.000089 | 0.892192 | 0.919748 | 0.911934 |
| **Latency** | 11.891212 | 0.18460271633512396 | 0.027263 | 11.694424 | 12.088238 | 11.970417 |

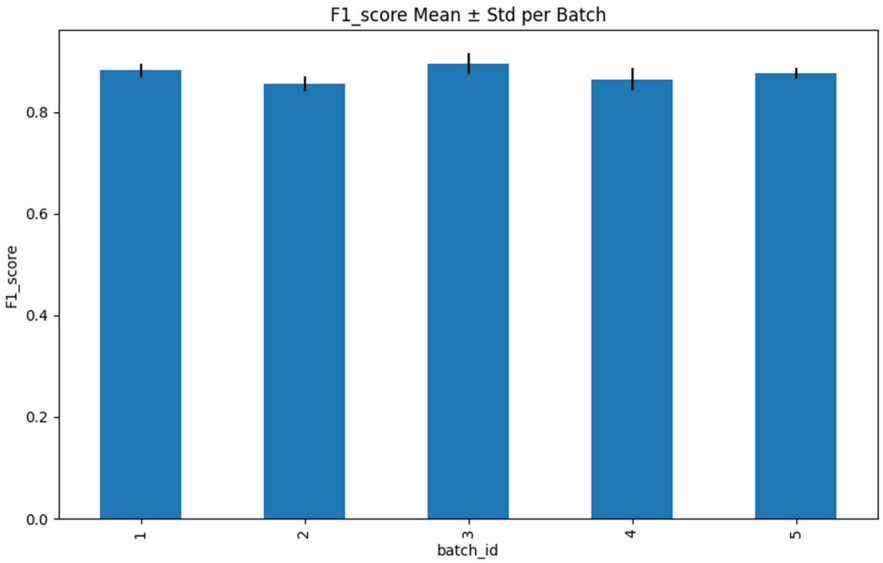
# 4. Correlation Analysis

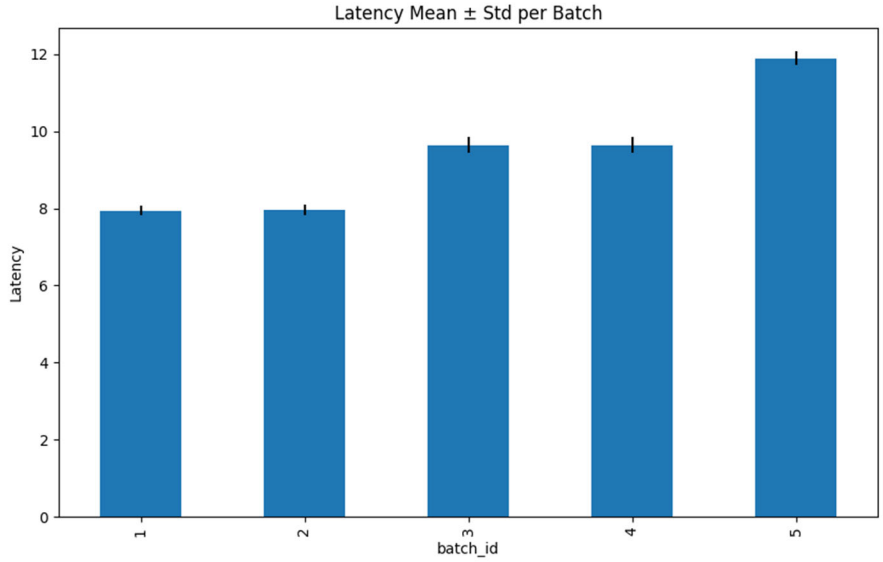


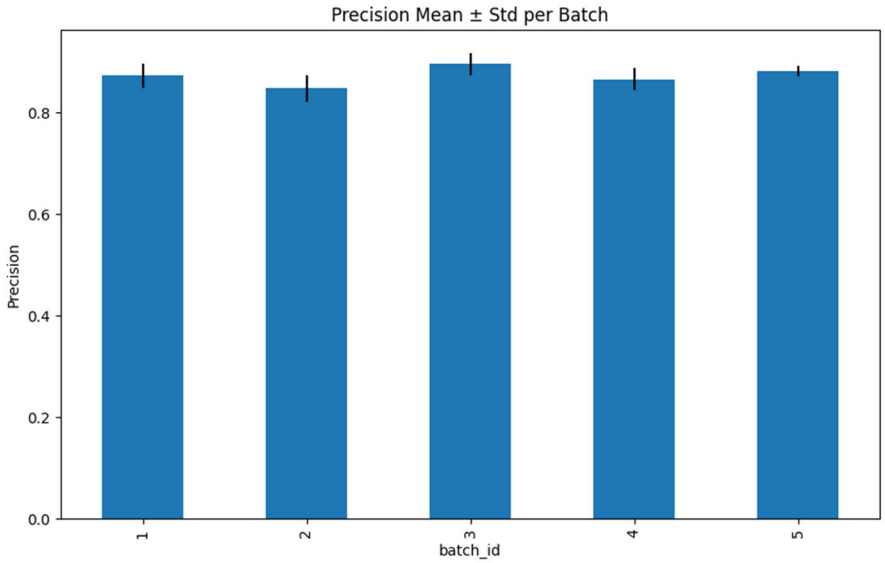
# 5. Visualizations

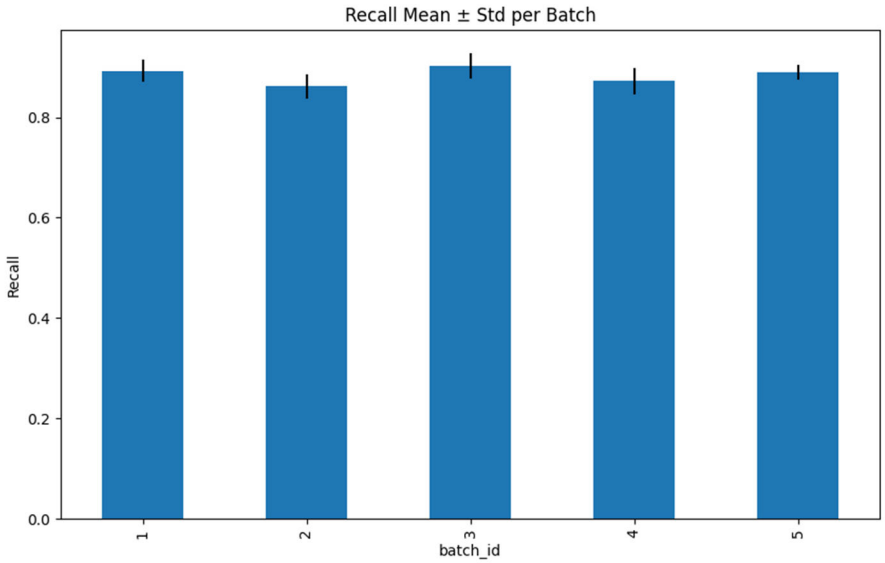


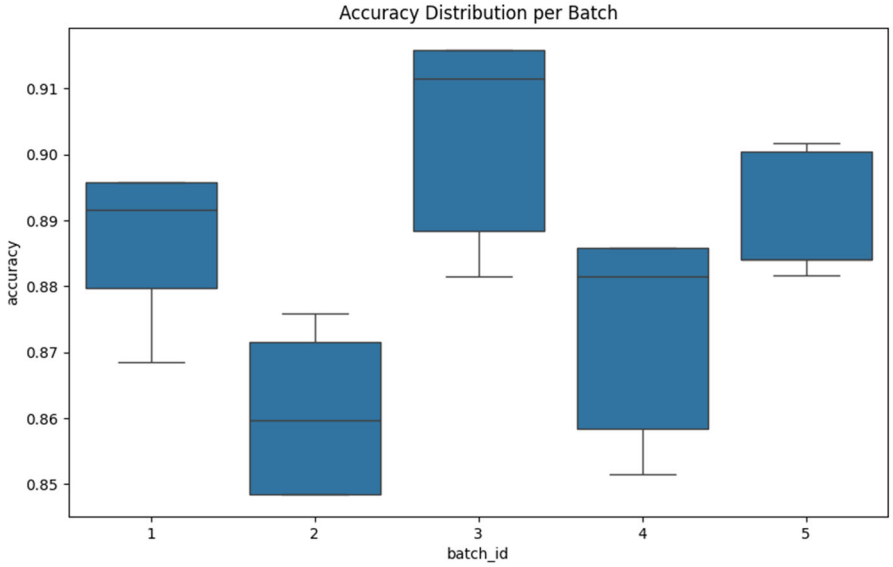



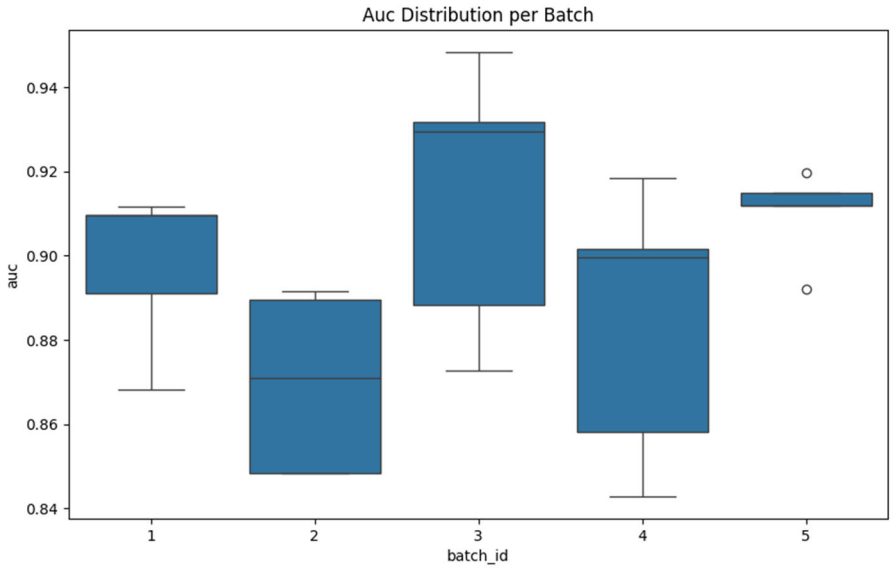

F1_score Mean ± Std per Batch
F1_score
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

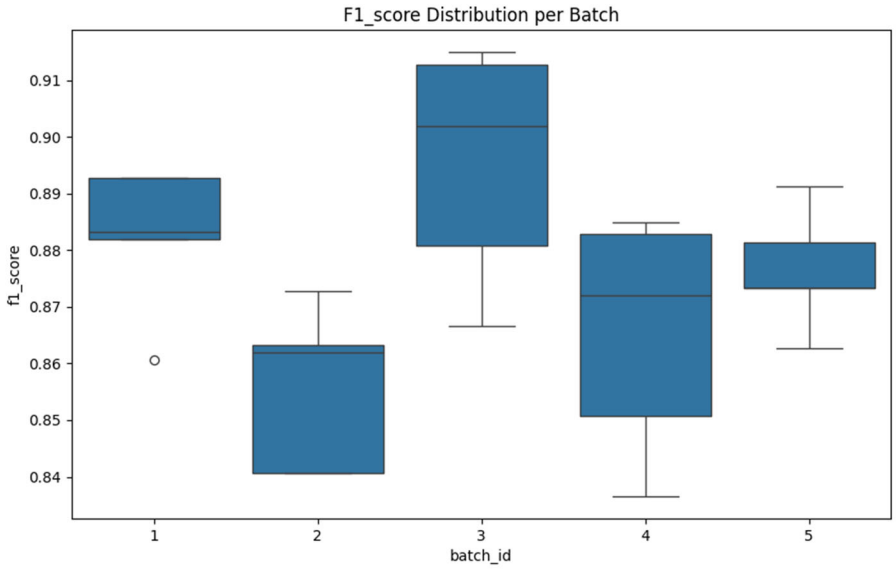

Latency Mean ± Std per Batch
Latency
0
2
4
6
8
10
12
1
2
3
4
5
batch_id

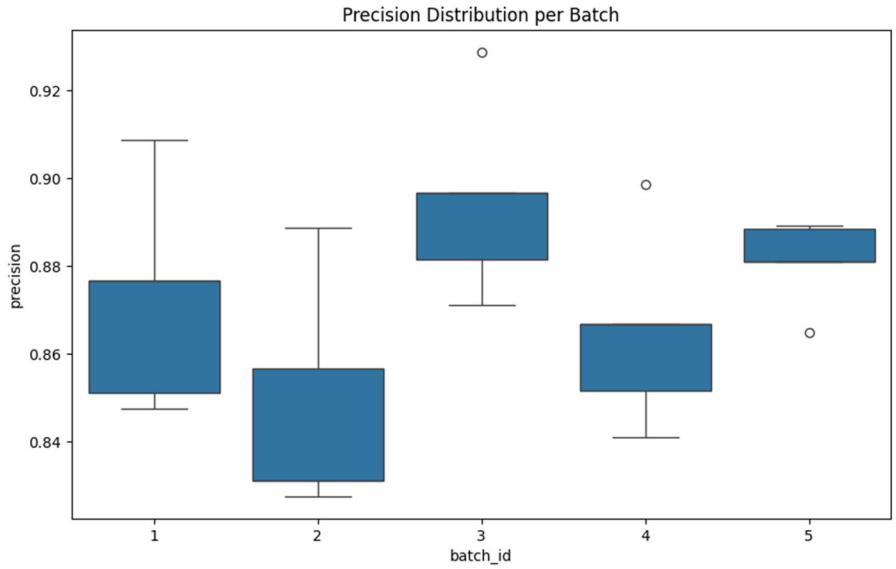

Precision Mean ± Std per Batch
Precision
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

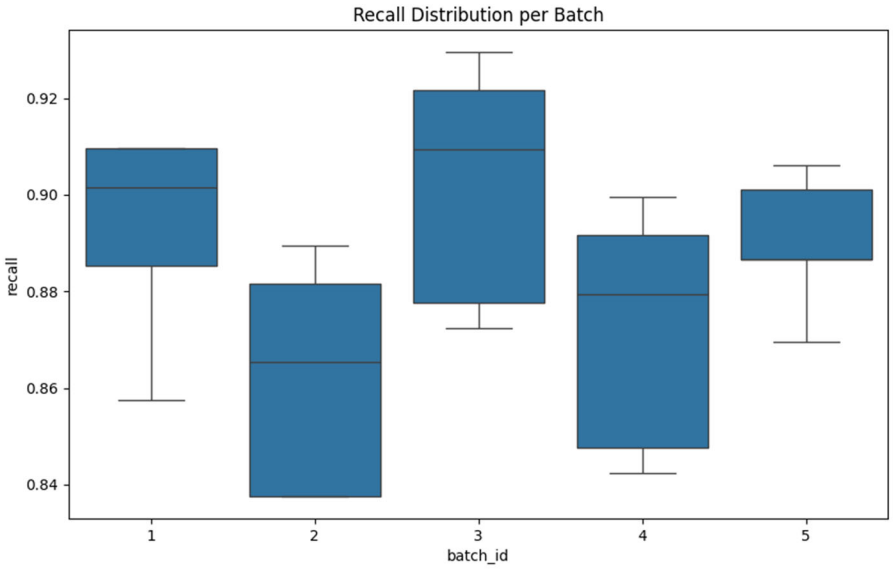

Recall Mean ± Std per Batch
Recall
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

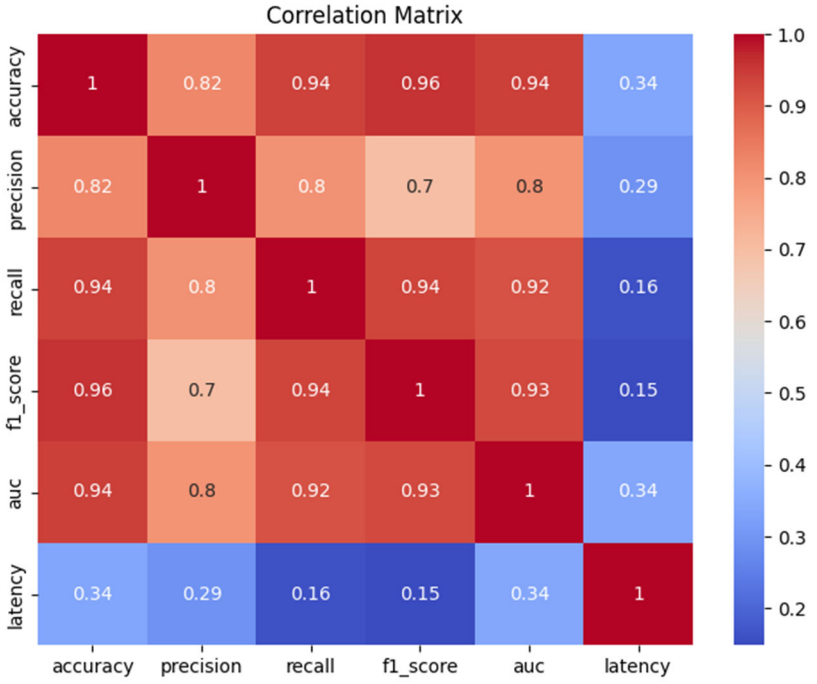

Accuracy Distribution per Batch
accuracy
0.91
0.90
0.89
0.88
0.87
0.86
0.85
1
2
3
4
5
batch_id

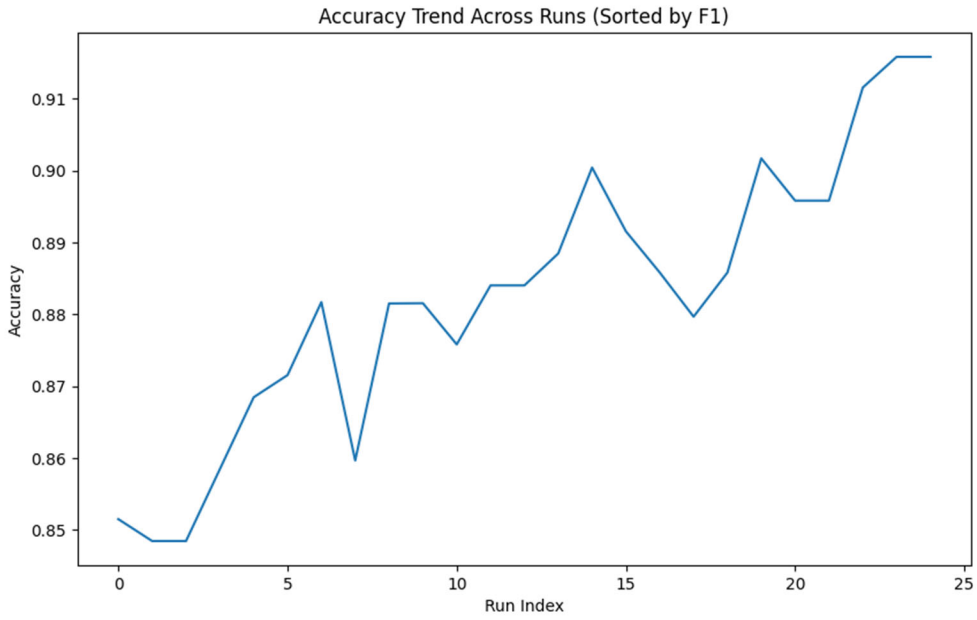

Auc Distribution per Batch
auc
0.94
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5
batch_id

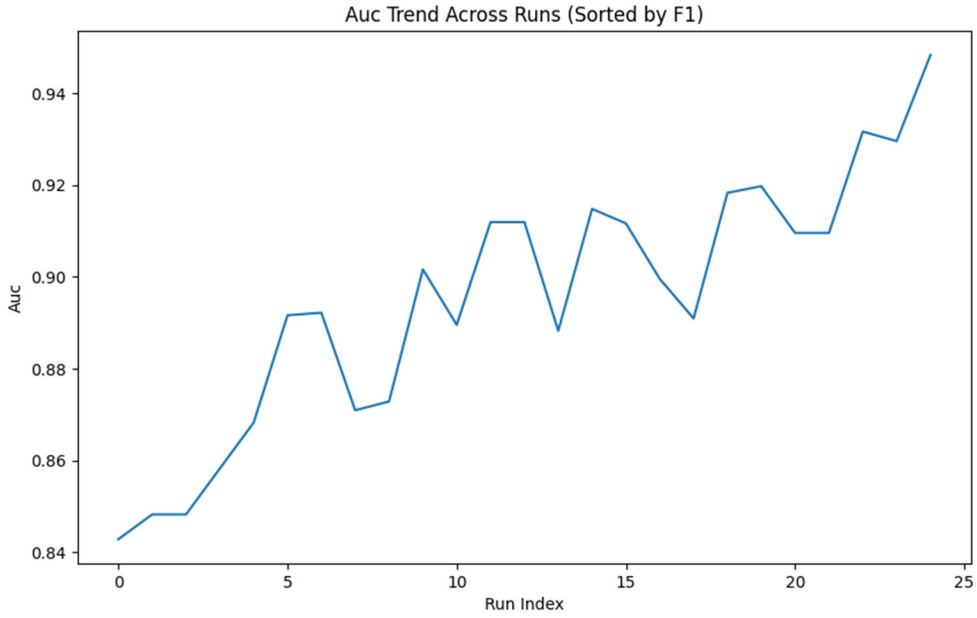

F1_score Distribution per Batch
f1_score
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.84
1
2
3
4
5
batch_id

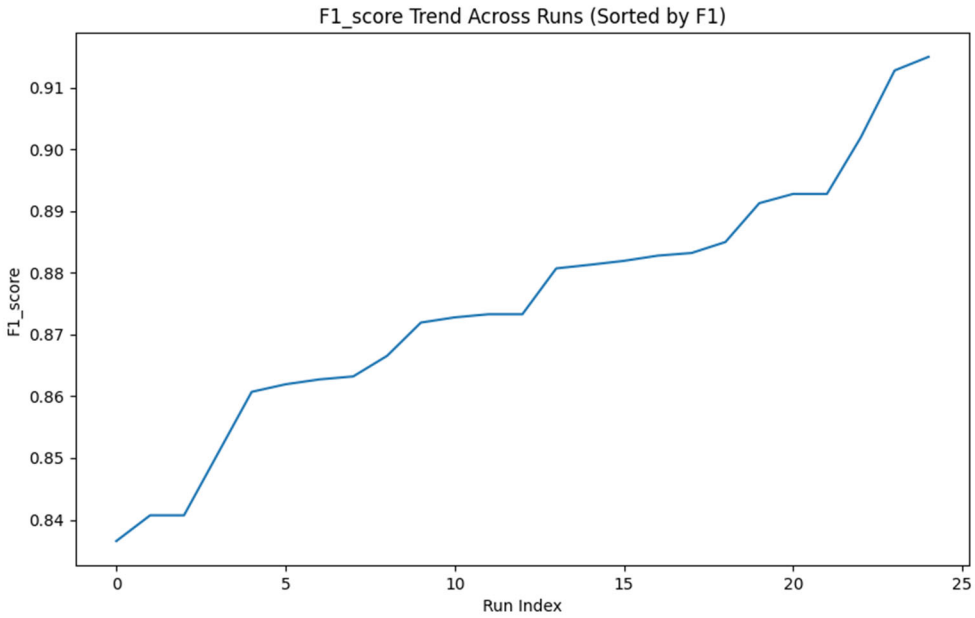

Precision Distribution per Batch
precision
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5
batch_id

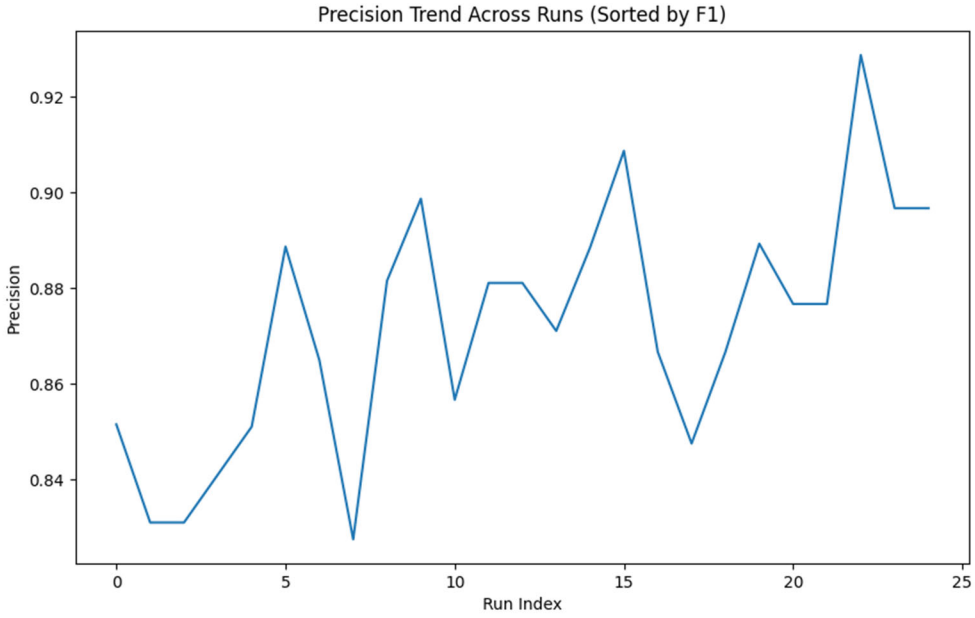

Recall Distribution per Batch
recall
batch_id
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5

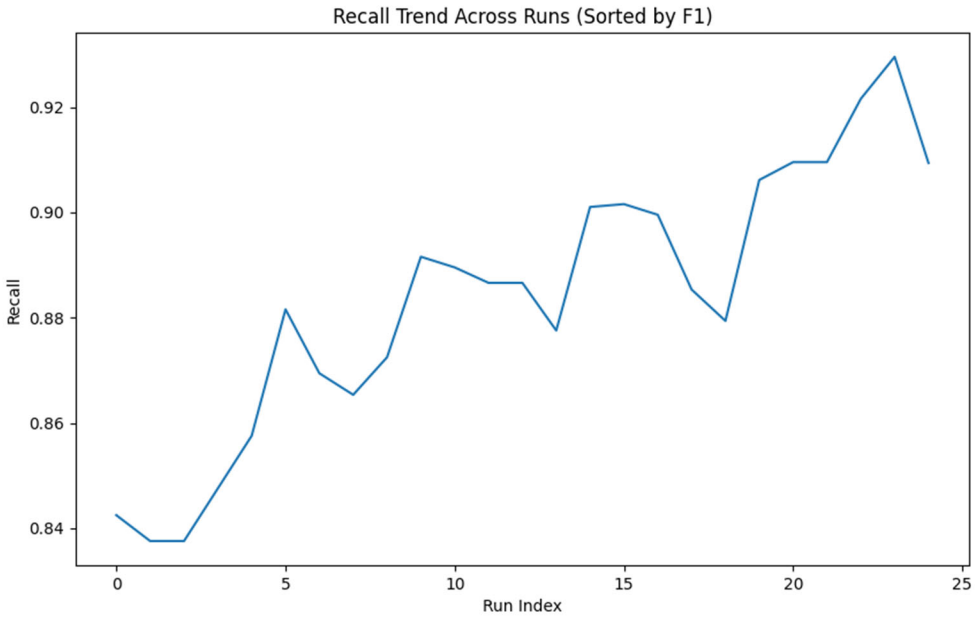

Correlation Matrix
accuracy
precision
recall
f1_score
auc
latency
1 0.82 0.94 0.96 0.94 0.34
0.82 1 0.8 0.7 0.8 0.29
0.94 0.8 1 0.94 0.92 0.16
0.96 0.7 0.94 1 0.93 0.15
0.94 0.8 0.92 0.93 1 0.34
0.34 0.29 0.16 0.15 0.34 1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2


Accuracy Trend Across Runs (Sorted by F1)
Accuracy
Run Index
0.85
0.86
0.87
0.88
0.89
0.90
0.91
0
5
10
15
20
25


Auc Trend Across Runs (Sorted by F1)
Auc
Run Index
0.84
0.86
0.88
0.90
0.92
0.94
0
5
10
15
20
25


F1_score Trend Across Runs (Sorted by F1)
F1_score
Run Index
0.84
0.85
0.86
0.87
0.88
0.89
0.90
0.91
0
5
10
15
20
25


Precision Trend Across Runs (Sorted by F1)
Precision
Run Index
0.84
0.86
0.88
0.90
0.92
0
5
10
15
20
25


Recall Trend Across Runs (Sorted by F1)
Recall
Run Index
0.92
0.90
0.88
0.86
0.84
0
5
10
15
20
25

## 6. Key Insights & Analysis

- Best Performing Metric: LATENCY with mean 9.411680
- Most Stable Metric: ACCURACY with std 0.018973
- Highest Variance: LATENCY with variance 2.129293
- Best Run: Batch 3 Run run_4 with F1 0.914991
- Worst Run: Batch 4 Run run_5 with F1 0.836546
- Total evaluation samples: 150 across 5 batches